

Problem

A **knowledge graph** (KG) is a network of facts stored as triples (**subject, relation, object**). **Link prediction** asks: given $(h, r, ?)$, which entity t is most likely to complete the triple?



Models learn by comparing correct triples against *negatives* (fabricated wrong triples). The quality of negatives is critical.

RANDOM NEGATIVES (STANDARD)

Obama → born.in → *Ice cream*

Obama → born.in → *Mt. Everest*

Obama → born.in → *Pacific Ocean*

Too easy. Model stops improving.

HARD NEGATIVES (OURS)

Obama → born.in → *Chicago*

Obama → born.in → *Kansas City*

Obama → born.in → *Honolulu*

Plausible but wrong. Model keeps improving.

Honolulu is the correct answer here, making it the hardest possible negative for this query.

Motivation

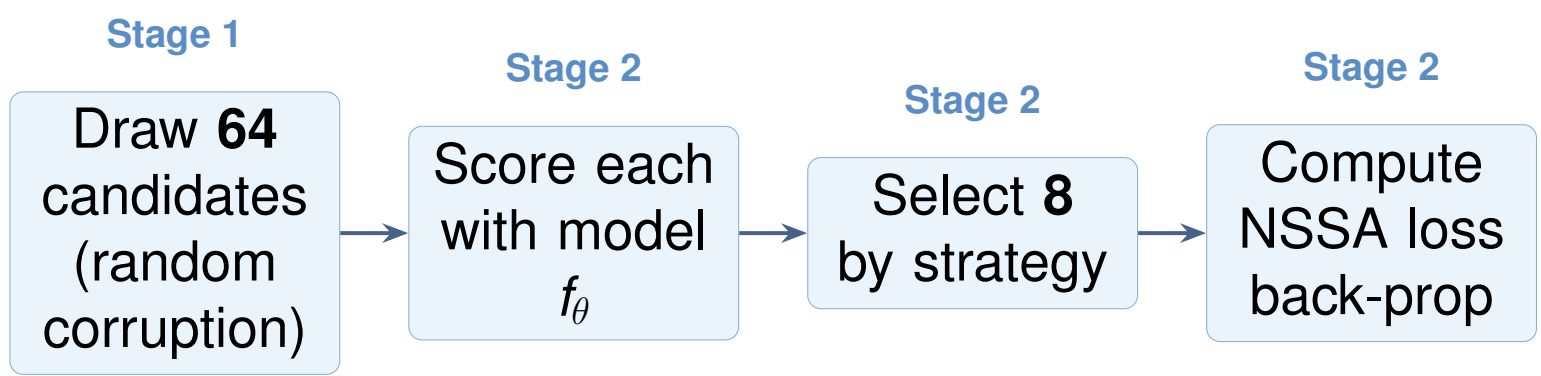
Random negatives become trivial after a few epochs: the model rejects them easily, so the **gradient signal vanishes** and learning plateaus.

Hard negatives keep the model challenged throughout training, but using them exclusively can destabilise learning. **A mixed strategy** combines both.

Question: What mixture gives the best accuracy–stability trade-off on FB15k-237?

Methodology

For each positive triple at training time:



Selection strategies tested in Stage 2:

- **Random:** sample 8 uniformly (control)
- **Hard:** take the 8 highest-scoring wrong candidates
- **Mixed** p : take fraction p hard + $(1 - p)$ random, with $p \in \{30\%, 50\%, 70\%\}$

All strategies share the same Stage 1 pool: zero extra compute.

Common setup: RotatE, embedding dim 128, batch 1024, Adam lr= 10^{-3} , 50 epochs (patience 10), seed 42.

References

[1] Sun *et al.* (2019). *RotatE*. ICLR 2019.
[2] Wang *et al.* (2014). *Knowledge Graph Embedding by Translating on Hyperplanes (Bernoulli sampler)*. AAAI 2014.
[3] Toutanova & Chen (2015). *FB15k-237 dataset*. ACL Workshop 2015.

Datasets

FB15k-237 [3] — Freebase benchmark, widely used for KG link prediction.
Entities: 14,505 Train triples: 272,115
Relation types: 237 Val/ Test: 17,526/20,438
Relation frequencies span **three orders of magnitude** (37–15,989 occurrences), enabling stratified slice evaluation.

Results

+10.4 %

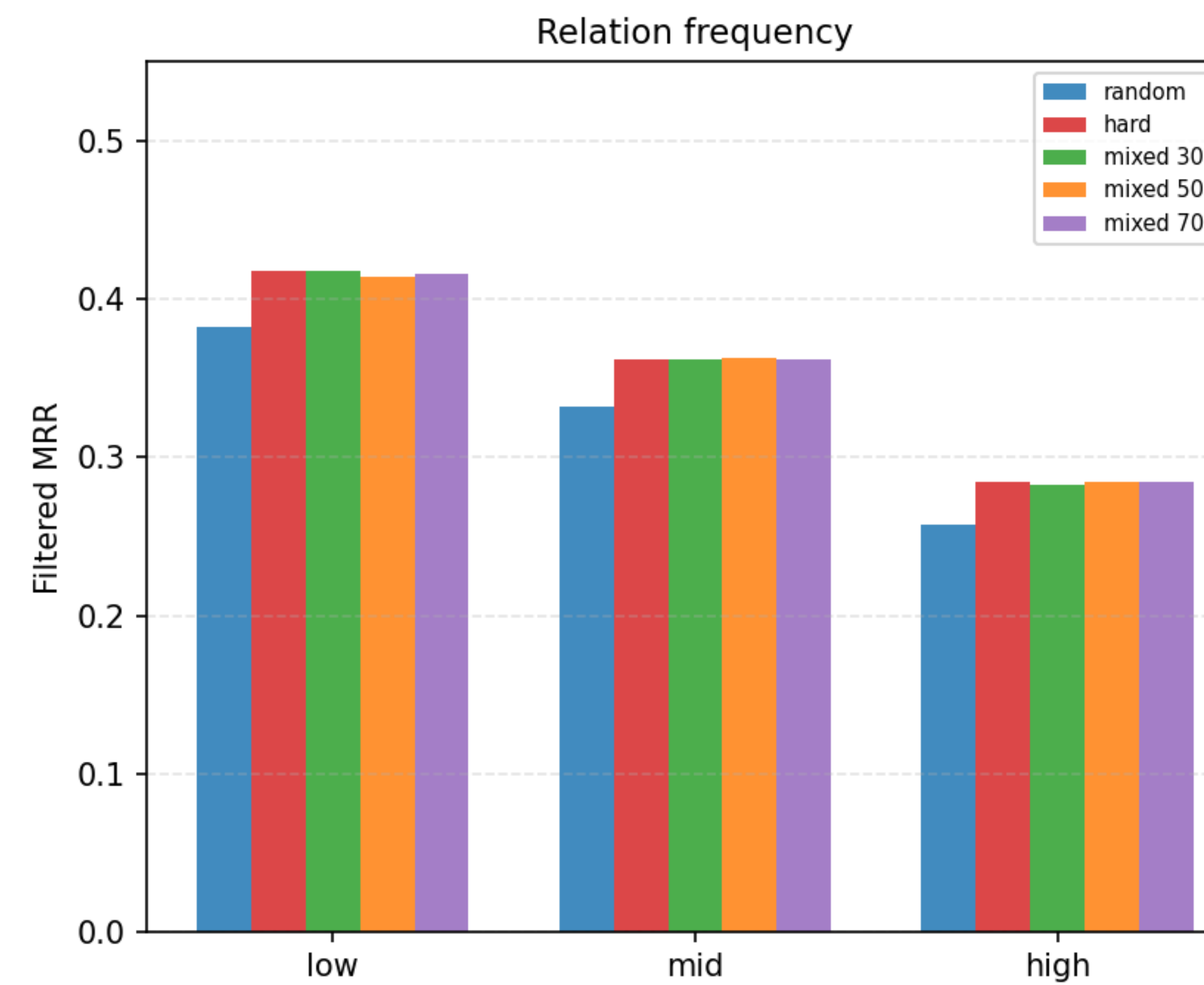
accuracy gain (MRR) with Hard negatives vs. standard random sampling

Strategy	MRR	H@1	H@3	H@10
Random (baseline)	0.270	0.189	0.300	0.428
Hard	0.298	0.217	0.327	0.459
Mixed 30%	0.297	0.215	0.326	0.459
Mixed 50%	0.298	0.216	0.328	0.460
Mixed 70%	0.299	0.217	0.328	0.461
Δ Hard – Random	+0.028	+0.028	+0.026	+0.031

Hard or mixed negatives improve ranking by +10.4%, especially on rare and difficult relation / entity slices.

Mixed 30% hard recovers 95% of the gain at zero extra cost.

Slice evaluation — MRR by relation frequency (test set, filtered):



Gain is largest on **rare relations** (+3.6 pp: $0.382 \rightarrow 0.418$) where random negatives are least informative.

Qualitative example — query: (Film, release.region, ?), gold: **France**

Strategy	Rank of France	
Random (control)	9	buried
Hard	1	✓ correct
Mixed 30%	1	✓ correct
Mixed 50%/70%	3	near top

Hard training pushes France from rank 9 to rank 1. Even 30% hard fully recovers it.

Conclusions

- **Hard negatives are a free upgrade:** same training pipeline, smarter selection step, +10.4% MRR gain.
- **You don't need 100% hard:** Mixed 30% (2 hard examples out of 8) recovers 95% of the gain with added stability.
- **Rare and ambiguous facts benefit most**, which is exactly what KGs are built for.